

Agriculture Advisory Assistant

Fine-Tuning a Small LLM for Real-World Agricultural Q&A

Group Members

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1. Problem Definition

1.1 Use Case

Smallholder farmers in sub-Saharan Africa often lack access to timely, localized agricultural advice. Extension services are under-resourced and internet connectivity is unreliable. This project addresses the gap by building a lightweight, instruction-tuned language model capable of answering practical farming questions in English, covering crop management, soil health, irrigation, and pest control.

The assistant targets contexts where a small, deployable model is preferable to a large API-dependent system. The use case is intentionally narrow and well-defined: a farmer or field agent types a question in plain language and receives a concise, actionable answer.

1.2 Target Users

- Smallholder farmers seeking quick advice on crop disease, fertilizer, or irrigation.
- Agricultural extension agents who need a rapid reference tool in the field.
- NGO workers and development practitioners supporting farmers with limited connectivity.

1.3 Why Fine-Tuning Is Needed

General-purpose language models are not optimized for domain-specific agricultural Q&A. They tend to produce verbose, generic, or regionally inappropriate answers. Fine-tuning on curated agricultural datasets allows the model to produce concise, domain-relevant responses aligned with the format and depth expected by the target audience.

Fine-tuning also enables instruction-following behavior to be shaped to a specific task format (instruction + category → assistant answer), which is more predictable than prompting a general model with complex system messages at runtime.

1.4 Model choice

Compute constraints are real in the deployment context. The model must run inference at a reasonable cost and potentially be hosted on a free-tier Hugging Face Space. Qwen3-0.6B was selected because it offers a strong instruction-following baseline at a scale that fits within limited VRAM, trains quickly with LoRA, and can be served efficiently with quantization.

2. Data Preparation

2.1 Data Sources

To build the Agriculture Assistant, three complementary datasets were selected from the Hugging Face Hub. Each dataset contributes a different type of knowledge to improve the model's ability to answer agricultural questions.

Source Name	HuggingFace ID	Type	Used Examples	Category
source_1_general	Satyam66/Agricare	General agricultural instruction	10000	agriculture_general
source_2_synthetic	KisanVaani/agriculture-qa-english-only	Synthetic QA pairs	5000	synthetic
source_3_domain	talhakk/agriculture-qa	Domain expert QA	5000	agriculture_domain

2.2 Data Schema

The selected datasets use different schemas and therefore required normalization before training.

talhakk/agriculture-qa

Field	Description
question	User question
answer	Corresponding answer

KisanVaani/agriculture-qa-english-only

Field	Description
question	User question
answers	Generated answer

Satyam66/Agricare

Field	Description
instruction	Agricultural instruction or question
input	Additional context (optional)
output	Expected answer

2.3 Cleaning and Filtering

- Removed rows with empty instruction or output fields.
- Deduplicated examples with identical instruction strings (case-insensitive).
- Stripped HTML tags and whitespace artefacts from raw dataset outputs.
- Capped output length at 512 tokens to prevent outlier sequences dominating training.
- Assigned explicit source and source_detail fields for provenance tracking.

2.4 Data Formatting

Since the three datasets have different structures, they were converted into a unified instruction-following format.

Intermediate Format

Each example was transformed into a common schema:

```
{  
  "instruction": "...",  
  "input": "...",  
  "output": "...",  
  "source": "general | synthetic | domain"  
}
```

Chat Format

The instruction dataset was then converted into a conversational format compatible with Qwen3 fine-tuning:

```
{
  "messages": [
    {
      "role": "system",
      "content": "You are an expert agricultural assistant. Provide accurate, practical, and concise answers."
    },
    {
      "role": "user",
      "content": "<instruction>"
    },
    {
      "role": "assistant",
      "content": "<output>"
    }
  ]
}
```

This format enables supervised instruction tuning using the assistant-only training strategy, where the loss is computed only on the assistant responses.

3. Splitting Strategy

Splits were performed per source family with a fixed seed (seed=42) to ensure reproducibility. Each source was split independently before concatenation, preventing any single source from dominating the evaluation set.

Split	Ratio	Approx. Total Rows	Purpose
train	90%	18000	LoRA adapter optimization
validation	5%	1000	Loss monitoring during training
eval	5%	1000	Final automatic metrics

The split was performed separately for each data source to preserve the contribution of the general, synthetic, and domain-specific datasets.

4. Training Methodology

4.1 Base Model

The base model is Qwen/Qwen3-0.6B, a 0.6 billion parameter causal language model from Alibaba. It was selected for its strong instruction-following capabilities at small scale, compatibility with the transformers and PEFT libraries, and suitability for LoRA fine-tuning without full-precision training requirements.

4.2 LoRA Configuration

Hyperparameter	Value	Rationale
lora_r	16	Low-rank approximation; balances expressiveness and parameter count
lora_alpha	32	Scaling factor; $\alpha/r = 2$ is a standard starting point
lora_dropout	0.05	Light regularization to prevent adapter overfitting
target modules	q_proj, v_proj	Query and value projections; sufficient for instruction tuning
trainable params	~1.7M	Less than 0.3% of total model parameters

4.4 Training Configuration

Parameter	Value
Epochs	3
Batch size	1
Gradient accumulation	8 (effective batch = 8)
Learning rate	1e-4
Warmup steps	50
Weight decay	0.01
Max sequence length	512
Optimizer	AdamW (default)

4.5 Checkpointing and Monitoring

- Checkpoints saved every 500 steps; up to 10 checkpoints retained.

- Best adapter saved to outputs/.../best_adapter/ based on validation loss.
- Training curves monitored via TensorBoard (tensorboard --logdir outputs/.../runs).
- Checkpoint manager in src/checkpoint_manager.py handles cleanup automatically.

5. Evaluation

Evaluation is run on the held-out eval split (data/splits/eval_allx.jsonl) using evaluation.py with the --generate flag to perform actual model inference. The following metrics are reported:

Metric	Score	Notes
Exact Match	0.1350	Strict character-level match
Token F1	0.4125	Bag-of-words overlap at token level
BLEU	0.2697	n-gram precision with brevity penalty
ROUGE-L	0.3675	Longest common subsequence recall

Note: we evaluate on only 200 examples because of delay

6. Deployment

6.1 Hugging Face Hub

Model repository: Mumphone/agriculture-assistant-qwen3-lora

The LoRA adapter was pushed to the Hub using push_to_hub.py with --checkpoint best and --public flags. The repository contains only the adapter weights and configuration, not the full base model.

6.2 Hugging Face Space

Space URL: https://huggingface.co/spaces/Mumphone/agriculture_assistant

The Space is built with Gradio and loads the adapter from the Hub at startup, merging it with the base model at inference time. The interface provides a category selector and a free-text instruction field.

6.3 Guardrails Implemented

- Out-of-scope detection: prompts unrelated to agriculture trigger a refusal message.
- Minimum confidence check: if the context state machine returns no relevant examples, the app warns the user that the answer may be unreliable.

7. Limitations and Risks

7.1 Hallucination

As with all generative models, the assistant can produce plausible-sounding but factually incorrect agronomic advice. This is particularly dangerous in a domain where wrong advice (e.g., incorrect pesticide dosage, wrong planting depth) can cause crop loss or harm. The deployed app includes a disclaimer note on all responses.

7.2 Category Coverage Gaps

The training data covers crop management, soil health, irrigation, and pest control well, but has limited coverage of livestock, post-harvest storage, and market pricing. Queries in these areas may produce generic or off-topic responses. This was identified during qualitative evaluation.

7.3 Data Bias

The three source datasets were created primarily from English-language sources targeting South Asian and East African contexts. Advice may be regionally inappropriate for West African soil types, rainfall patterns, or locally available inputs.

7.4 Out-of-Scope Prompts

The model was not trained to refuse unsafe or out-of-scope requests. It may attempt to answer questions about topics it was never trained on, producing low-quality outputs. The deployed Space includes a basic keyword-based guardrail, but this is not a substitute for RLHF-based safety fine-tuning.

8. Improvements

Future improvements include expanding the agricultural dataset with more diverse and region-specific sources, integrating Retrieval-Augmented Generation (RAG) to access up-to-date agricultural information, and adding multilingual support. Further evaluation with agricultural experts and the development of advanced features such as voice interaction and personalized farming recommendations could also enhance the system's usefulness and reliability.

9. Team contributions

Team Member	Main Contributions
Lieumo NJOPNU	data preprocessing, fine-tuning, pipeline building, deployment and report writing
Joel SANGURA	Dataset collection, data processing, fine-tuning, deployment and report writing
Ahmed Souleymane SOW	Dataset collection, data preprocessing, evaluation experiments and report writing
Lekounda Gedeon	pipeline building, Hugging Face deployment, testing, evaluation, and report writing
Daouda SENE	model testing, result analysis, and report writing